English – Moroccan Darija (in Arabizi) Machine Translation via a Transformer Architecture

Ilias Bourhrara
Engineering Student
CentraleSupélec
Email: ilias.bourhrara@student-cs.fr

***Abstract*—Moroccan Darija is an Arabic-descended language, characterized by the absence of a standardized grammar and orthography, making its machine translation particularly complex. This work presents the development of an English–Darija *Transformer*-based machine translation model, optimized under a strict 100-million parameter constraint to enable offline execution on smartphones. We leverage a dataset of 280k sentence pairs generated via distillation from *Gemini 2.5 Flash*. To overcome the high lexical variability of Darija, we implement a single shared-vocabulary English-Darija *tokenizer*, coupled with *BPE Dropout* regularization. The model, trained from scratch, contains a total of 98M parameters. Given the absence of standard *BLEU* metrics suited for non-standardized languages, evaluation is conducted using LLM judgments, demonstrating remarkable performance against the language's structural challenges.**

1. Introduction

Machine translation into low-resource or non-standardized languages represents a major challenge for natural language processing. Moroccan Darija (Moroccan Arabic) falls squarely into this category: it is an essentially oral language, characterized by an absence of strict grammatical and orthographic norms. While typically written with the Arabic alphabet, we focus here on a direct translation into *Arabizi* (Latin characters), as a simple transliteration approach does not work (for example, transliteration translates *car* to "twmwbl" rather than "tomobil"). The bulk of online resources in this language and alphabet comes from digital communications, which has allowed large language models to assimilate it well.

The objective of this project is to design a high-performing English-to-Darija translation system while adhering to a strong hardware constraint: the final model must not exceed the 100-million parameter threshold, to guarantee its viability and smooth execution directly on smartphones.

2. Dataset Constitution and Normalization

The main obstacle to developing a model for Darija is the lack of sufficiently large parallel corpora, with the exception of the DODa project, which nevertheless contains few samples written in *Arabizi*.

2.1. Generation by Distillation

To overcome this lack of resources, we implement knowledge distillation from the *Gemini 2.5 Flash* model, which has a strong command of this language. This method allowed us to generate a high-quality parallel corpus comprising approximately 280K English–Darija sentence pairs. Extracts from this dataset are provided in the appendix. The dataset was partitioned conventionally: **95% for training** and **5% for validation**. This corpus remains modest in size, considering that *datasets* of several million sentences are typically required to train high-performing translation models.

2.2. Preprocessing and Sequence Management

The normalization of textual data mainly consists of converting the entire corpus to lowercase. The maximum sequence length was set to **256 tokens**, which generally proves sufficient for translating everyday sentences.

3. Tokenization Strategy

The orthographic variability of Darija constitutes a major hurdle for classical tokenization algorithms (e.g., *bezzaf*, *bezaf*, or *bzaf*, all meaning "a lot"). This is why the *tokenizer* proved to be, throughout the project, the main bottleneck limiting the model's progress.

3.1. Limitations of the Disjoint Approach

An initial attempt relied on the use of two distinct *BPE* tokenizers: an English tokenizer with a 32k-token vocabulary, and a Darija tokenizer with a 48k-token vocabulary. This method quickly showed its limitations by treating orthographic variants as completely independent entities. To illustrate, the model might, for example, hesitate among the three spellings *bezzaf*, *bezaf*, or *bzaf*, which leads it to dilute their probabilities and, ultimately, to generate another semantically disconnected token.

3.2. Optimization via Shared Vocabulary and BPE Dropout

To solve this issue, we opted for a single, shared tokenizer combining English and Darija, with a vocabulary size restricted to **32k tokens**. This approach favors better sub-lexical decomposition because the smaller tokenizer performs fewer merges:

$$\text{b—ez—zaf} \approx \text{b—e—zaf} \approx \text{b—zaf} \quad (1)$$

Furthermore, introducing the ***BPE Dropout*** technique during training forces the model to encounter different possible segmentations for the same word (for example, alternately `b|ez|zaf` or `bez|zaf`), thereby reinforcing its robustness against spelling variations. An example of corpus segmentation is illustrated below:

```
kanbghi| nt|m|acha| f| zzna|9|i| dyal|
                    l|7|ay|
```

4. Model Architecture and Training

The model is based on the standard *Transformer Encoder-Decoder* architecture but incorporates critical structural modifications to improve its convergence and efficiency.

4.1. Architectural Modifications

Weight Tying: The weight matrices for input and output *embeddings* are shared. This not only reduces the overall number of parameters but also improves training stability. Indeed, without *weight tying*, the output linear layer tends, during early iterations, to project massively toward the most frequent tokens, causing a collapse of representations that subsequently requires significant time to regularize.
Shared Vocabulary: Since the tokenizer used is common to both languages, we implemented a single *embedding* matrix, shared between the decoder and encoder, drastically reducing the amount of parameters.
Pre-Layer Normalization: Unlike the original *Post-LN* architecture, normalization is applied upstream of each sub-layer (*attention* and *feed-forward*) as well as at the output of the encoder and decoder, rather than on the residual path, which greatly stabilizes gradient flow.
RMSNorm: Traditional *LayerNorm* normalization was replaced by *RMSNorm*, which demonstrates equivalent performance while halving the computational cost.

4.2. Hyperparameters

To best exploit the imposed size limit, the model was dimensioned using the hyperparameters presented in Table 1.

The asymmetrical depths of the encoder and decoder enable better performance on a language as orthographically noisy as Darija. A high *dropout* rate (0.2) was also applied in the residual layers and attention matrices to compensate for the small *dataset* size and limit overfitting.

Table 1. HYPERPARAMETERS OF THE SELECTED ARCHITECTURE

Hyperparameter	**Value**
Encoder layers	10
Decoder layers	12
Embedding dimension (d_{model})	512
Attention heads per layer (n_{head})	8
Max Sequence Length	256 tokens
Total Parameters	**98 Million**

4.3. Training Configuration

The loss function used is the ***Cross Entropy Loss***, configured with a ***Label Smoothing*** of 0.1 to prevent the model from becoming overly confident when faced with the various legitimate spellings of the same word. Optimization relies on ***AdamW*** with a ***Weight Decay*** coefficient of 0.01. Gradients are bounded via ***Gradient Clipping*** to a maximum norm of 1.0 to prevent gradient explosion. Finally, a ***Learning Rate Scheduler*** combining a linear *warmup* followed by ***Cosine Annealing*** decay ensured optimal convergence, with a maximum learning rate set to 4.10^{-4}.

5. Inference

During the inference phase, token selection relies on a ***Frequency Penalty*** of 0.6 to penalize pattern repetitions. We explored classical generation strategies: *Top-K sampling*, *Top-P (Nucleus) sampling*, and *Beam Search*, the latter allowing us to maximize the overall coherence of the translated sentences. Decoding via ***Beam Search*** coupled with the *Frequency Penalty* was ultimately selected for final evaluations.

6. Evaluation and Future Work

The absence of standard automated evaluation benchmarks (such as certified *BLEU* test corpora) for Darija constitutes an inherent limitation for non-standardized languages. To circumvent this obstacle, evaluation was performed by submitting the translations produced by our model (98M parameters) to an external LLM to verify their semantic fidelity. In this instance, the *Gemini 3.5 Pro* model was used to validate the fluency and coherence of the predictions. The results obtained are highly promising, demonstrating an excellent capture of the language's nuances and even of certain idiomatic expressions.

Future work will primarily focus on the handling of proper nouns. Since these are often out-of-vocabulary terms (absent from the training corpus), they currently constitute one of the major difficulties for the model.

This work paves the way for generalization to other languages with non-standardized orthography and grammar, suffering from a deficit of online resources. For some of these languages, massive multilingual models such as *NLLB* (Meta) or *T5* (Google) have already demonstrated remarkable capabilities that could inspire future compressed architectures.

7. Appendices

7.1. Observation of Trained Embeddings

To visualize the semantic encoding performed by the model during training, a two-dimensional principal component analysis (PCA) was applied to the latent space of the *embeddings*.

Figure 1 highlights the natural formation of semantic clusters grouping concepts by similar themes, demonstrating a logical and structured organization of the representation space.

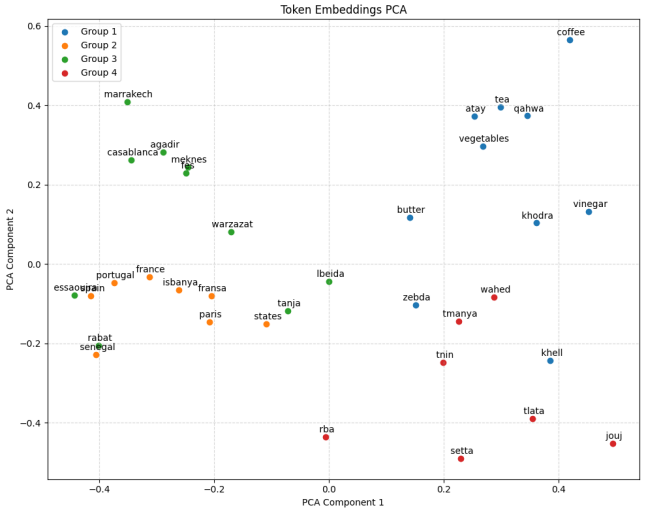


Figure 1. PCA visualization of thematic *clusters* within the *embedding* space.

Furthermore, cross-lingual alignment was also verified. As illustrated in Figure 2, the projections reveal that identical concepts in both languages cluster in close proximity, experimentally validating the effectiveness of the single tokenizer and shared vocabulary strategy.

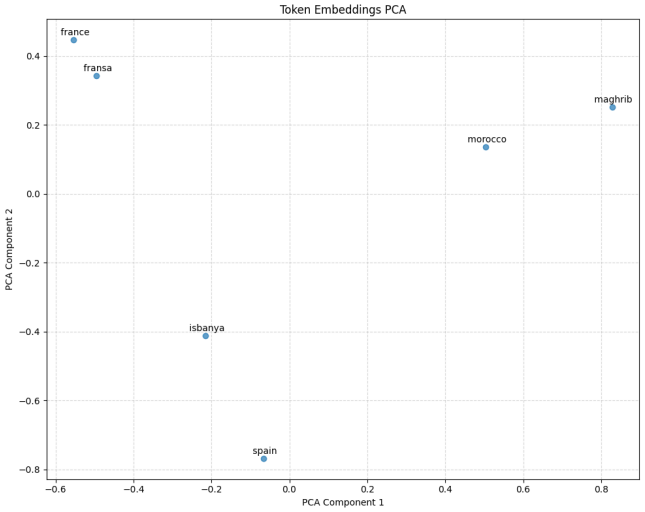


Figure 2. Geometric alignment of the same semantic concepts (countries) across the two languages.

7.2. Dataset Extracts

The dataset built via distillation includes sentence pairs of varying lengths. Table 2, located on the following page, presents examples of short and long extracts from the English–Darija parallel corpus.

7.3. Generated Translation Examples

The generalization capabilities of the 98M-parameter model were evaluated on complex statements with strong cultural or contextual weight, some results of which are presented in Table 3 on the following page.

Table 2. EXTRACTS OF SHORT AND LONG EXAMPLES FROM THE TRAINING DATASET

English (Source)	**Moroccan Darija (Target)**
Examples of short sentences	
I'm not familiar with the route to Rabat, so I'll rely on your experience. Could you ensure we take the quickest and safest way?	Ma 3arfch triq l Rabat, dakchi 3lach ghan3tamed 3la khibrtek. Momkin tdamen liya nakhdo asra3 w a2man triq?
My budget is limited, I can't pay more than 55 dirhams.	Lbudget diali m7doud, ma nqderch nkhallas kter men 55 dirham.
Example of a long sentence	
The international community's increasing recognition of the Sahara's intrinsic link to Morocco's historical and cultural identity, alongside its strategic importance for regional security and stability, means that any resolution must fully integrate the specificities of this unique region within the broader framework of Moroccan sovereignty, ensuring that the local populations are empowered to manage their affairs while remaining an integral part of the Kingdom, thus securing a peaceful and prosperous future.	L2i3tiraf lli kaykber mn lmujtama3 ddawli b rraba6 l2asassi dyal ssahra b lhowiya ttariikhiya w ttakafiya dyal lmghrib, m3a l2ahmiya listratijiya dyalha l2amn w listi9rar ljiihawi, kay3nii bali ay 7all khassu yddmaj bchakl kamil lkhousousiyat dyal had lmin6a9a lfaaridda dakhil l2i6ar l3am dyal ssiyada lmaghribiya, kayaddamman tamkin nnas lma7alliyin bach yssayro chomounhom m3a lba9a2 kjjuz2 la yatajaza2 mn lmamlakka, w bttali kayaddamman musta9bal ssalamii w raakhii.

Table 3. EXAMPLES OF TRANSLATION OUTPUTS OBTAINED AT INFERENCE

Source	**Model Translation**
Can you show me the way to the old medina? I'm completely lost and my phone's battery is dead.	wach t9der twerini tri9 l lmedina l9dima? ana telft kaml w tele dyali ma b9ach fih la batterie.
I spent nearly forty-five minutes this morning arguing with the merchant at the weekly souk over the price of fresh mint and orange blossom water needed for the family gathering, entirely forgetting that my grandmother expected me back hours ago to help roll the semolina grains for the Friday couscous, which meant I had to hurriedly catch a shared grand taxi under the blazing midday heat just to avoid her legendary disapproval.	dwezzt chi khamsa w arba3in dqiqa had sba7 wana kan3awd m3a lbiya3 f souq l'ousbou3i 3la taman dyal na3na3 jdid w ma zhar li khasso l jam3a dyal la famille, nssit ga3 belli jaddati kant katssennani men qbel swaye3 bach n3awen ndawwer 7boub smida l couscous dyal nhar ljomu3, dakchi li khasni khassni nchadd taxi kbir mouchtarak ta7t skhouna dyal nos nhar ghir bach netfadda rafdha l'os6ori.